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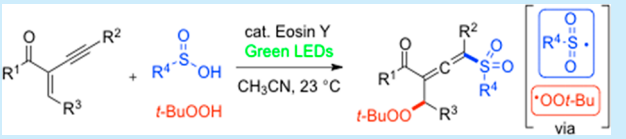
Photoredox-Catalyzed 1,4-Peroxidation−Sulfonylation of Enynones: A Three-Component Radical Coupling Approach for the Synthesis of Highly Functionalized Allenes

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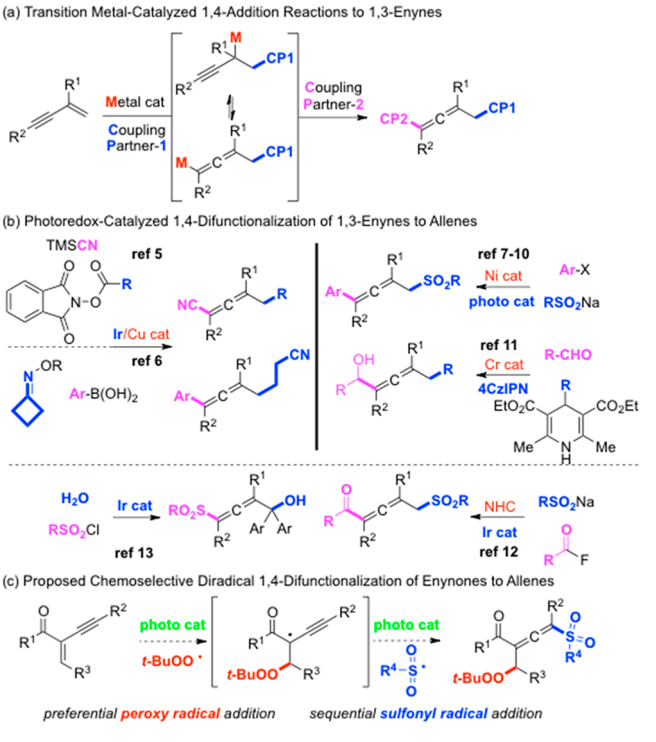
ABSTRACT: An Eosin Y-catalyzed visible light-promoted 1,4-peroxidation−sulfonylation of enynones was achieved to give tetrasubstituted allenes. The photoredox catalysis of Eosin Y allowed the concomitant formation of peroxy and sulfonyl radicals, where the preferential peroxy radical addition to the alkene moiety of enynones resulted in the subsequent α-keto radical−sulfonyl radical cross couplings. The developed photoredox catalysis of Eosin Y demonstrates a regioselective 1,4-diradical addition strategy, opening up a new possibility of diradical functionalization of conjugate systems.



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Given the widespread synthetic utility of cumulenes,[1] the synthesis of functionalized allenes such as allenols[2] and allenones[3] has attracted considerable interest from the synthetic community. As one of the direct allenylation strategies,[4] the transition metal-catalyzed addition reactions to 1,3-enynes provide the metalated allenyl intermediates that lead to the related allenyl products under thermodynamic control (Scheme 1a). The metalated allenyl intermediate was first utilized by the Lu's group to give CN-substituted allenes under the visible light-induced dual Ir-photoredox/Cu-catalyzed 1,4-carbocyanation of 1,3-enynes (Scheme 1b).[5] The use of cyclobutanone oxime esters as alkyl radical precursors was explored by the Guo's group under similar dual Ir-photoredox/Cu-catalyzed conditions to give the highly functionalized allenes.[6] The regioselective 1,4-arylsulfonylations of 1,3-enynes were independently developed by the groups of Li,[7] Wang,[8] and Lu[9] under the dual catalysis of photoredox and nickel, where sulfinate salts and aryl halides were used as coupling partners. In 2022, the Kong's group also disclosed the combination of photo-hydrogen atom transfer catalyst, decatungstate, and nickel catalysis for the 1,4-dicarbofunctionalization of 1,3-enynes to give tetrasubstituted CF_3-allenes.[10] More recently, the Wang's group developed the dual 4-CzIPN photoredox and chromium catalysis for the asymmetric 1,4-difunctionalization of 1,3-enynes to the chiral allenols.[11] In contrast to the previous dual metal/photoredox catalysis approaches, the Zhang's group explored the dual Ir-photoredox/NHC-catalyzed 1,4-sulfonylacylation of 1,3-enynes using sodium sulfinate salts and aroyl fluorides.[12] Such a nonmetal-based 1,4-hydroxysulfonylation of 1,3-enynes was also explored by the Wu's group under the Ir-photoredox catalysis using water and sulfonyl chlorides.[13] Inspired by the recent advances in the visible light-promoted photoredox sulfonylation[14] as well as the metal-catalyzed radical perox-

Scheme 1. 1,4-Difunctionalization of 1,3-Enynes and the Proposed Photoredox Catalysis Approach to 1,4-Peroxidation−Sulfonylation



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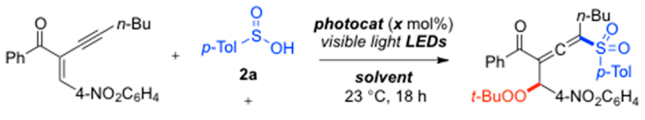
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idation,[15] we became interested in a visible light photoredox catalysis for the 1,4-peroxidation/sulfonylation of 1,3-enynes (Scheme 1c). Our group recently reported the chemodivergent sulfonylation of enynones, where the sulfonyl radical addition occurred at the alkene moiety of enynones.[16] Nevertheless, the prediction of the reaction pathways of two different radical species, peroxy and sulfonyl, was deemed to be challenging because of the persistent character of peroxy and sulfonyl radical species. Herein, we report the visible light photoredox catalysis of Eosin Y for the diradical functionalization of enynones, ensuring the chemoselective peroxy radical addition, followed by a consequent addition of the sulfonyl radical to enynones. The developed visible light photoredox diradical 1,4-functionalization of 1,3-enynes provides highly functionalized allene derivatives via the sequential installation of two different radical species in a three-component coupling strategy.

The photoredox-catalyzed 1,4-difunctionalization of 1,3-enynes was explored using enynone **1a**, *p*-toluenesulfinic acid **2a**, and *t*-butyl hydroperoxide (TBHP in 70 wt.% in H_2O) in CH_3CN under visible light LEDs (Table 1). The use of Ir- and Ru-based photoredox catalysts led to the formation of desired allene **3a** as a diastereomeric mixture in 8−15% yields (entries 1 and 2). The screening of organo-photocatalysts revealed that the formation of **3a** could be improved to 29−44% yields upon using Rose Bengal (entry 4), Mes-AcrBF$_4$ (entry 5), Na_2-eosin Y (entry 7), and Eosin Y (entry 8). The subsequent solvent screening under Eosin Y catalysis did not fare well (entries 9−14), only a mixed solvent of DCE:H_2O showed a marginally comparable outcome to the 37% yield of **3a** (entry 10). The increased catalyst loading of Eosin Y did little to improve the reaction efficiency (entry 15). Eventually, the significantly improved yield of **3a** was achieved upon increasing the TBHP content to 6 equiv (entry 17), where the product **3a** was isolated in 68% yield. The optimal use of 3 mol % Eosin Y photocatalyst was needed (entry 18). The subsequent variation of visible light source confirmed the necessity of green LED's for the Eosin Y photocatalysis (entries 19−21). The control experiments in the absence of Eosin Y photocatalyst (entry 22) or visible light (entry 23) confirmed the critical needs of visible light LED's and Eosin Y photocatalyst for the formation of **3a**. The reaction under air instead of an argon atmosphere negatively influenced the formation of **3a** in 15% yield, suggesting the decomposition pathways of the involved radical species under an oxygen atmosphere (entry 24). The use of a nonaqueous solution of TBHP in decane also provided the desired product **3a** in 64% yield (entry 25), a slightly lower yield than the TBHP in water (entry 17).

With the optimized visible light Eosin Y catalysis for the 1,4-peroxidation−sulfonylation of enynone, the substrate scope was investigated using various enynones **1** and sulfinic acids **2** (Scheme 2). The electronic and steric effects of the ketone moiety (R^1) of enynones **1** were investigated using phenyl groups with electron-donating (**3b**) and electron-withdrawing groups (**3c** and **3d**), as well as cyclohexyl (**3e**) and 2-furanyl (**3f**) groups, where the desired tetrasubstituted allenes were obtained as 1:1−1:1.8 diastereomeric mixtures in 37−63% yields. The R^2 substituent of the alkyne moiety in enynones **1** tolerated alkyl (**3g**), alkyl chloride (**3h**), ester (**3i**), and *N*-phthaloyl (**3j**) groups to give the desired products in 32−60% yields. The R^3 substituent of the alkene moiety in enynones **1** could accommodate the electronically diverse phenyl and heteroaryl groups to give tetrasubstituted allenes **3k**−**3o** in 36−59% yields. The radical coupling partner, sulfinic acid **2**, could also tolerate electronically diverse phenyl sulfinic (**3p**−**3r**) and aliphatic sulfinic groups (**3s** and **3t**).

Table 1. Optimization of Visible Light Eosin Y-Catalyzed 1,4-Peroxidation-Sulfonylation of Enynone[a]



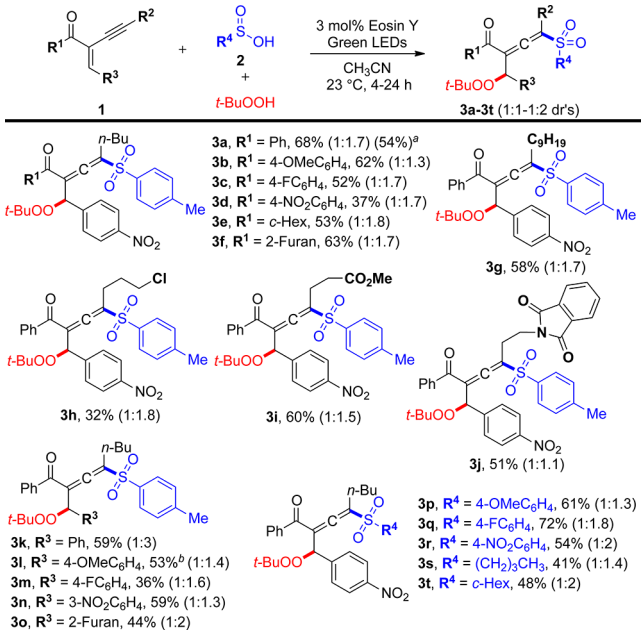
entry	photocat (mol %)	LEDs	solvent	% yield[b] (dr)
1	*fac*-Ir(ppy)$_3$ (3)	purple	CH_3CN	15 (1:1.6)
2	Ru(bpy)$_3$Cl$_2$ (3)	blue	CH_3CN	8 (1:4.0)
3	methylene blue (3)	blue	CH_3CN	11 (1:4.0)
4	Rose Bengal (3)	green	CH_3CN	31 (1:1.5)
5	Mes-AcrBF$_4$ (3)	purple	CH_3CN	41 (1:1.4)
6	fluorescein (3)	purple	CH_3CN	15 (1:1.4)
7	Na_2-eosin Y (3)	green	CH_3CN	29 (1:1.8)
8	Eosin Y (3)	green	CH_3CN	44 (1:1.8)
9	Eosin Y (3)	green	DCE	22 (1:1.6)
10	Eosin Y (3)	green	DCE:H_2O	37 (1:1.3)
11	Eosin Y (3)	green	$PhCH_3$	11 (1:1.4)
12	Eosin Y (3)	green	acetone	17 (1:1.6)
13	Eosin Y (3)	green	DMSO	0
14	Eosin Y (3)	green	MeOH	0
15	Eosin Y (5)	green	CH_3CN	38 (1:2.0)
16[c]	Eosin Y (3)	green	CH_3CN	54 (1:1.8)
17[d]	Eosin Y (3)	green	CH_3CN	72[e] (1:1.6)
18[d]	Eosin Y (1)	green	CH_3CN	62 (1:1.9)
19[d]	Eosin Y (3)	blue	CH_3CN	39 (1:1.6)
20[d]	Eosin Y (3)	purple	CH_3CN	25 (1:1.8)
21[d]	Eosin Y (3)	white	CH_3CN	32 (1:1.6)
22[d]		green	CH_3CN	13 (1:2.0)
23[d]	Eosin Y (3)		CH_3CN	11 (1:1.5)
24[d,f]	Eosin Y (3)	green	CH_3CN	15 (1:1.8)
25[d,g]	Eosin Y (3)	green	CH_3CN	64 (1:1.2)

[a]Reaction conditions: **1a** (0.3 mmol), **2a** (0.45 mmol), TBHP (0.9 mmol in 70 wt % in H_2O), photocat (mol %) in solvent (0.1 M) under visible light LEDs under argon at 23 °C for 18 h. [b]Yields determined by ^{1}H NMR using an internal standard (diastereomeric ratio of **3a** in parentheses). [c]TBHP (1.35 mmol, 4.5 equiv) was used. [d]TBHP (1.8 mmol, 6.0 equiv) was used in 10 h. [e]Isolated yield of **3a** in 68%. [f]Reaction under air. [g]TBHP in decane (5.5 M, 1.8 mmol, 6.0 equiv).

To better understand the current visible light photocatalysis of Eosin Y, the 1,4-peroxidation−sulfonylation of enynones **1a** was performed in the presence of radical inhibitors (Scheme 3). The desired product **3a** was not observed upon adding 2 equiv of TEMPO, only leading to the formation of the sulfinic acid addition product **4a** in an ionic mode.[16] However, the HRMS analysis of the crude reaction mixture confirmed the detection of TEMPO-O*t*-Bu **4b** and TEMPO-captured allenic or propargyl adducts **4c**, suggesting the possible generation of the *t*-butoxy radical and propargyl radical species. While the use of another radical capture reagent, butylated hydroxytoluene (BHT), failed to intercept the radical species, 1,1-diphenylethylene provided the corresponding sulfonyl- and *t*-butyl peroxy radical-captured products **4d** and **4e** in 12−16% yields. The hydrogen atom of sulfinic acid **2** turned out to be critical, since the sodium sulfinate **2a-Na**, instead of sulfinic acid **2a**, led to the exclusive formation of the alkynyl epoxide **5a** and **5b** in 54−61% yields. In such an epoxidation pathway,

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Scheme 2. Substrate Scope of Visible Light Eosin Y-Catalyzed 1,4-Peroxidation−Sulfonylation of Enynones

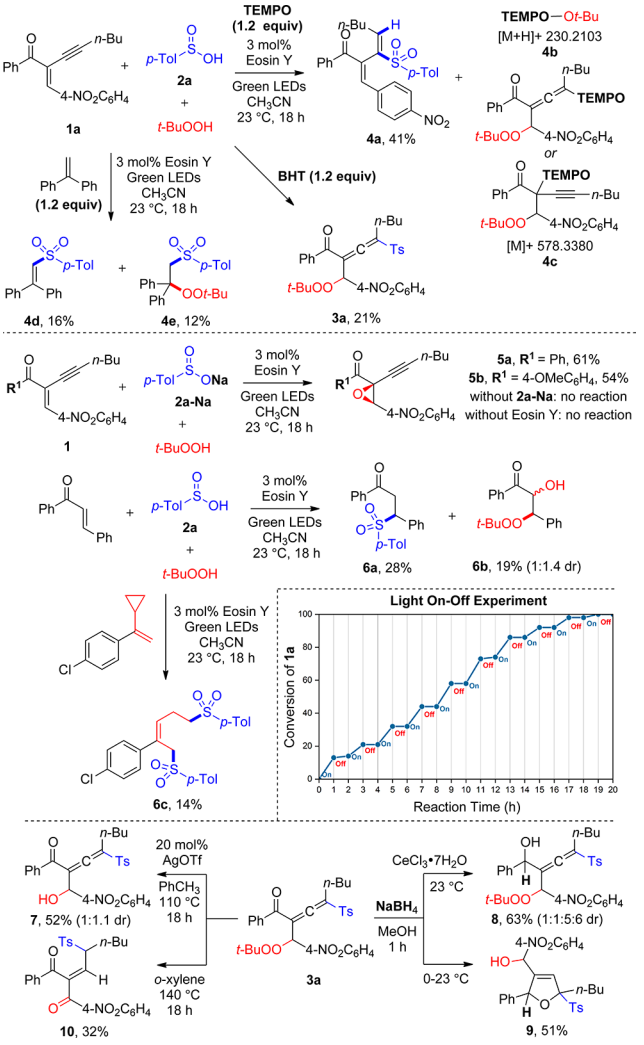


[a]Reaction in 3 mmol. [b]Reaction in DCE:H_2O (4:1).

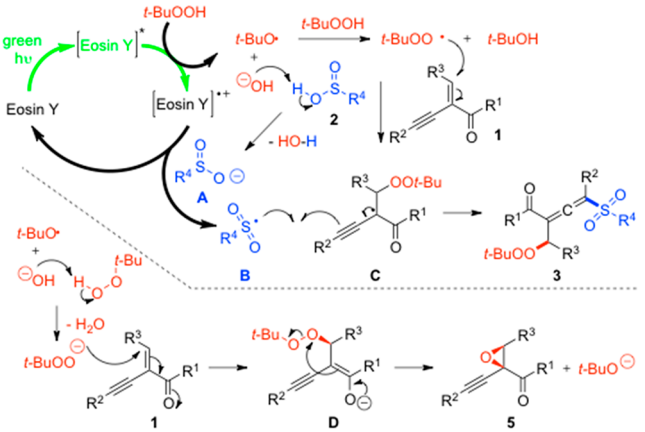
each reaction component, sodium sulfinate **2a-Na**, TBHP, Eosin Y photocatalyst, and green LEDs, played important roles in the reaction; thus, the omission of any component led to no reaction. The employment of chalcone, instead of **1a**, also provided sulfonyl addition product **6a** and *t*-butyl peroxy addition product **6b** in 28% and 19% yields, respectively. The radical clock experiment confirmed the radical addition to the alkene moiety, where the adjacent cyclopropyl ring was opened and the preferential sulfonyl addition product **6c** was obtained in 14% yield. The fluorescence quenching experiments indicated that the excited state of Eosin Y was quenched by THBP,[18b] while the sulfinic acid **2a** could not quench the Eosin Y (see the Supporting Information for details). The light on/off experiment suggested the necessity of continued visible light irradiation for meaningful reaction progress. The peroxide moiety in the allene **3a** could be manipulated to give the allenol **7** in 52% yield under the AgOTf catalysis. Alternatively, the ketone moiety of **3a** could be selectively reduced to the corresponding alcohol **8** in 63% yield using $CeCl_3$−$NaBH_4$ reduction conditions, whereas the concomitant cyclization to the product **9** was observed in 51% yield using $NaBH_4$ alone. Upon heating in *o*-xylene, the disproportionation of **3a** was observed to give the 2-alkylidene-1,3-diketone **10** in 32% yield along with 4-nitrobenzaldehyde as a byproduct.

Based on the control experiments, a plausible reaction mechanism of the photoredox catalysis of Eosin Y for 1,4-peroxidation−sulfonylation of enynones is proposed in Scheme 4. With the maximum absorption of Eosin Y around 530 nm,[17] the application of green LED's generates the photoexcited state of [Eosin Y]* that subsequently undergoes a single electron transfer to TBHP to give a *t*-butoxy radical and a hydroxyl anion.[18] The subsequent hydrogen abstraction by the hydroxyl anion from the sulfinic acid **2** (pK_a = 7.1; TBHP pK_a = 12.8) is believed to be critical since the use of sodium sulfinate **2a-Na** led to the epoxidation pathway of enynones **1**. The resulting sulfinate anion **A** should undergo a single electron transfer to

Scheme 3. Control Experiments for Visible Light Photoredox Catalysis of Eosin Y and Product Derivatization



Scheme 4. Plausible Reaction Mechanism for Photoredox Catalysis of Eosin Y for 1,4-Peroxidation−Sulfonylation of Enynones



the Eosin Y radical cation $[Eosin]^{\bullet+}$ to give the sulfonyl radical **B** with the regeneration of the photocatalyst Eosin Y. The fate of the *t*-butoxy radical is governed by the ample presence of TBHP, where the formation of the *t*-butyl peroxy radical and *t*-BuOH should be encouraged.[19] A ready availability of the *t*-

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butyl peroxy radical species initiates the radical addition to the alkene moiety of enynones **1**, where the resulting α-keto radical species **C** can be stabilized by the adjacent carbonyl and alkynyl groups.[4] The final radical−radical coupling between the sulfonyl radical **B** and the α-keto radical **C** occurs at the γ-carbon of enynones **1** with less steric effect to give the allene product **3**. In the absence of the sulfinic acid **2** the Eosin Y-promoted photolytical formation of hydroxyl anion acts as a base to deprotonate TBHP. The resulting *t*-butyl peroxide anion should behave as nucleophile adding to the enynones **1** to provide the epoxide product **5** via the intermediate **D**.[20] It should be noted that while the previous photoredox-catalyzed reactions utilized TBHP as oxidants,[18a,b,21] the current results demonstrated the first peroxidation reaction involving TBHP under the photoredox catalysis.

In summary, we developed a visible light photoredox catalysis system for the 1,4-difunctionalization of enynones using a readily available organophotocatalyst, Eosin Y. The regioselective two different radical addition strategies were made possible by using TBHP and sulfinic acids, where the preferential *t*-butyl peroxy radical addition to the alkene moiety of enynones followed by the resulting α-keto radical-sulfonyl radical cross-coupling was achieved. With ready access to highly functionalized tetrasubstituted allenes, the developed visible light photoredox catalysis of Eosin Y should be applicable to the difunctionalization of other conjugated systems, 1,*n*-enynes, using two different radical species. Our current efforts are directed to such radical difunctionalization strategies, and these results will be subject of future publications.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its online Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.4c00517.

Experimental procedures and characterization data for all compounds and nuclear magnetic resonance (NMR) spectra (PDF)

FAIR data, including the primary NMR FID files, for compounds **3a**−**3t**, **4a**, **4d**, **4e**, **5a**, **5b**, **6a**−**6c**, **7**, **8**, **9**, and **10** (ZIP)

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Notes

The authors declare no competing financial interest.

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